

Optimization Techniques

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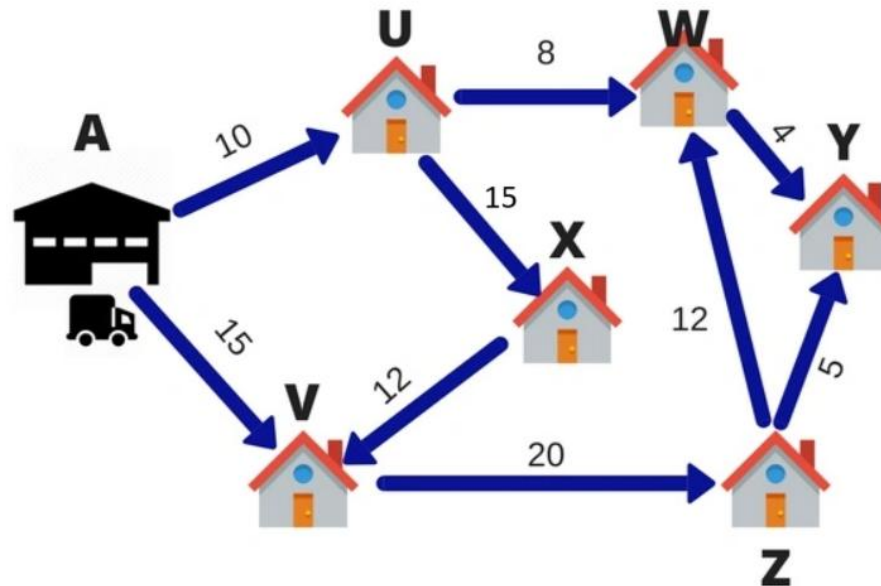
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What is Optimization?

Optimization is the process of finding the best possible solution to a problem from all available options, usually by maximizing or minimizing a specific objective.

Example 1:



Example:

You have 2 hours of study and want to prepare for **Math** and **English**.

- Studying **Math** gives you 5 marks per hour
- Studying **English** give you 3 marks per hour

You want to **maximize your total marks**.

Choice 1: 2 hr. **Math** $\Rightarrow 2 \times 5 = 10$ Marks

Choice 2: 1 hr. **Math** & 1 hr. **English** $\Rightarrow 5 + 3 = 8$ Marks

Choice 1: 2 hr. **English** $\Rightarrow 2 \times 3 = 6$ Marks

Best option: Study math for 2 hours to get maximum marks.

Mathematical Formulation of Optimization Problem (OP):

Let, $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $g_i: \mathbb{R}^n \rightarrow \mathbb{R}$, for $i \in \{1, 2, \dots, m\}$
 $h_j: \mathbb{R}^n \rightarrow \mathbb{R}$, for $j \in \{1, 2, \dots, p\}$ and $D \subset \mathbb{R}^n$.

The general form of a OP reads as:

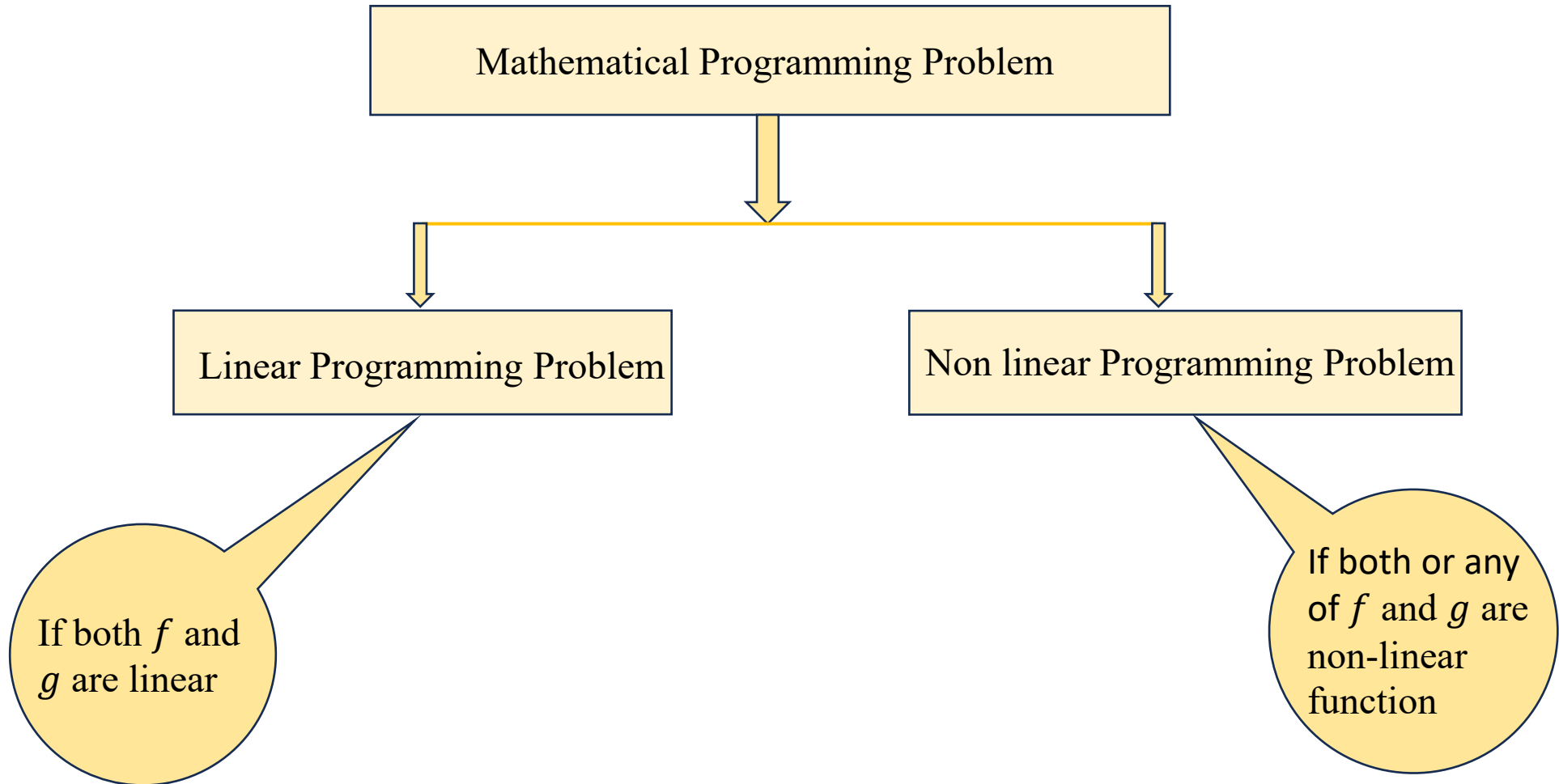
$$\text{Max or Min } f(x)$$

Subject to: (1.1)

$$\begin{aligned} g_i(x) &\leq 0, & \text{for } i \in \{1, 2, \dots, m\} \\ h_j(x) &= 0, & \text{for } j \in \{1, 2, \dots, p\} \\ x &\in D. \end{aligned}$$

- $x \in \mathbb{R}^n$ is the vector called decision variable.
- The function f is called objective function.
- The functions g and h are called constraint function.

Classifications:



Linear Programming Problem (LPP)

Formulation of LPP:

A small factory makes two product: **Tables** and **Chairs**.

- Each table require 4 hours of carpentry and 2 hours of painting.
- Each chair require 3 hours of carpentry and 1 hour of painting.
- The factory has 240 hours of carpentry and 100 hours of painting available in each week.
- Profit from each table is €70 and from each chair is €50.

Formulate a linear programming problem to determine the number of table and chairs that the factory should produce in each week to maximize the profit.

Solution:

Item	Carpentry (Hr.)	Painting (Hr.)	Profit (Eur.)
Table	4	2	70
Chair	3	1	50
Availability	240	100	

Let, x_1 tables and x_2 chairs should be produced in week.

Total profit in a week $Z = 70x_1 + 50x_2$

Total **Carpentry** time = $4x_1 + 3x_2$.

Total **Painting** time = $2x_1 + x_2$.

$$\text{Max } Z = 70x_1 + 50x_2$$

Subject to,

$$4x_1 + 3x_2 \leq 240,$$

$$2x_1 + x_2 \leq 100,$$

$$x_1, x_2 \geq 0.$$

Problem 2: A hotel operating 24 hours has the following minimum requirement of waiter:

Period	Time	Minimum No. of waiter required
1	7 am – 11 am	6
2	11 am – 3 pm	12
3	3 pm – 7 pm	8
4	7 pm – 11 pm	16
5	11 pm – 3 am	5
6	3 am – 7 am	3

A waiter reports to the hotel manager at the beginning of the period and continues to work for 8 hours. Formulate the LPP to minimise the total number of waiter required.

Solution:

Subject to,

$$\text{Min } Z = x_1 + x_2 + \dots + x_6$$

$$x_1 + x_2 \geq 12,$$

$$x_2 + x_3 \geq 8,$$

$$x_3 + x_4 \geq 16,$$

$$x_4 + x_5 \geq 5,$$

$$x_5 + x_6 \geq 3,$$

$$x_6 + x_1 \geq 6,$$

$$x_1, x_2, \dots, x_6 \geq 0 \text{ and integers.}$$

Problem 3: A furniture company manufactures desks and chairs. The sawing department cuts the lumber for both products, which is then sent to separate assembly departments. Assembled items are sent to the painting department for finishing. The daily capacity of the sawing department is 200 chairs or 80 desks. The chair assembly department can produce 120 chairs daily, and the desk assembly department 60 desks daily. The paint department has a daily capacity of either 150 chairs or 110 desks. Given that the profit per chair is \$50 and that of a desk is \$100, determine the optimal production mix for the company.

$$\text{Max } Z = 50x + 100y$$

Subject to,

$$4x + 10y \leq 800$$

$$x \leq 120$$

$$y \leq 80$$

$$11x + 15y \leq 1650$$

$$x, y \geq 0.$$

Problem 4: Reddy Mikks produces both interior and exterior paints from two raw materials, $M1$ and $M2$. The following table provides the basic data of the problem:

	Tons of raw material per ton of		Maximum daily availability (tons)
	<i>Exterior paint</i>	<i>Interior paint</i>	
Raw material, $M1$	6	4	24
Raw material, $M2$	1	2	6
Profit per ton (\$1000)	5	4	

The daily demand for interior paint cannot exceed that for exterior paint by more than 1 ton. Also, the maximum daily demand for interior paint is 2 tons.

Reddy Mikks wants to determine the optimum (best) product mix of interior and exterior paints that maximizes the total daily profit.

$$\text{Max } Z = 5x + 4y$$

Subject to,

$$6x + 4y \leq 24$$

$$x + 2y \leq 6$$

$$x - y \leq 1$$

$$y \leq 2$$

$$x, y \geq 0.$$

Problem 5: Ozark Farms uses at least 800 lb of special feed daily. The special feed is a mixture of corn and soybean meal with the following compositions:

Feedstuff	lb per lb of feedstuff		Cost (\$/lb)
	<i>Protein</i>	<i>Fiber</i>	
Corn	.09	.02	.30
Soybean meal	.60	.06	.90

The dietary requirements of the special feed are at least 30% protein and at most 5% fiber. The goal is to determine the daily minimum-cost feed mix.

Let x_1 lb of corns and x_2 lb of soyabeen meal need to be mixed.

$$\text{Min } Z = 0.3x_1 + 0.9x_2$$

Subject to,

$$x_1 + x_2 \geq 800$$

$$0.09x_1 + 0.60x_2 \geq \frac{30}{100}(x_1 + x_2)$$

$$0.02x_1 + 0.06x_2 \geq \frac{5}{100}(x_1 + x_2)$$

$$x_1, x_2 \geq 0.$$

Problem 6: (Trim-Loss Problem)

The Pacific Paper Company produces paper rolls with a standard width of 20 feet each. Special customer orders with different widths are produced by slitting the standard rolls. typical orders (which may vary daily) are summarized in the following table:

Order	desired Width(ft)	desired number of rolls
1	5	150
2	7	200
3	9	300

In practice, an order is filled by setting the knives to the desired widths. Find the minimum number of standard rolls to be slit according to the given data. Formulate the problem as an LPP model.

Knife setting

Required width	1	2	3	4	5	6	Min. No. of rolls
5	0	2	2	4	1	0	150
7	1	1	0	0	2	0	200
9	1	0	1	0	0	2	300
Trim loss per foot of length	4	3	1	0	1	2	

$$\text{Min } Z = 4x_1 + 3x_2 + x_3 + x_5 + 2x_6$$

Subject to,

$$2x_2 + 2x_3 + 4x_4 + x_5 \geq 150$$

$$x_1 + x_2 + 2x_5 \geq 200$$

$$x_1 + x_3 + 2x_6 \geq 300$$

$$x_1, x_2, \dots, x_6 \geq 0.$$

Problem 7:

A manufacturing company has a contract to deliver 100, 250, 190, 140, 220 and 110 home windows over the next six months. Production cost per window varies by period and is estimated to be \$50, \$45, \$55, \$48, \$52 and \$ 50 over next six months. To take advantage of fluctuations in manufacturing cost, company can produce more windows than needed in a given month and hold the extra units for delivery in later months. This will incur a storage cost at rate of \$8 per window per month. Formulate the problem as a linear programming problem.

Let x_i be the number of windows produced in the i th month $1 \leq i \leq 6$. and e_i be the extra units produced for i th month $1 \leq i \leq 6$.

Objective function: $\text{Min } Z = 50x_1 + 45x_2 + 55x_3 + 48x_4 + 52x_5 + 50x_6$

Subject to:

$$x_1 - e_1 = 100$$

$$(x_2 + e_1) - e_2 = 250$$

$$(x_3 + e_2) - e_3 = 190$$

$$(x_4 + e_3) - e_4 = 140$$

$$(x_5 + e_4) - e_5 = 220$$

$$(x_6 + e_5) - e_6 = 110$$

$$e_6 = 0,$$

$$x_i \geq 0, e_i \geq 0, \quad \text{for } 1 \leq i \leq 6.$$

Problem 8:

John must work at least 20 hours a week. To supplement his income while attending school, he has the opportunity to work in two retail stores. In store 1, he can work between 5 and 12 hours a week and in store 2, he can work between 6 and 10 hours. Both stores pay the same hourly wage. In deciding, how many hours to work in each store, John wants to base his decision on work stress, which are 8 and 6, respectively. Assume that total stress is proportional to total number of hours, determine how many hours John should work in each store?

Solution: **Decision Variables:** $\begin{cases} x_1 \text{ hours for store 1} \\ x_2 \text{ hours for store 2} \end{cases}$

Objective function: $\text{Min } Z = 8x_1 + 6x_2$

Subject to: $5 \leq x_1 \leq 12,$
 $6 \leq x_2 \leq 10, \quad x_1, x_2 \geq 0$

Problem 9:

A manufacturer produces three models, I, II, and III, of a certain product using raw materials *A* and *B*. The following table gives the data for the problem:

Raw material	Requirements per unit			Availability
	<i>I</i>	<i>II</i>	<i>III</i>	
<i>A</i>	2	3	5	4000
<i>B</i>	4	2	7	6000
Minimum demand	200	200	150	
Profit per unit(\$)	30	20	50	

The labor time per unit of model I is twice that of II and three times that of III. The entire labor force of the factory can produce the equivalent of 1500 units of model I. Market requirements specify the ratios 3:2:5 for the production of the three respective models. Formulate the problem as a linear program, and find the optimum solution.

Let, the manufacturer produces x_1 unites of Model I, x_2 unites of Model II and x_3 units of Model III product.

Total Profit: $\text{Max } Z = 30x_1 + 20x_2 + 50x_3$ (Objective function)

Subject to: $2x_1 + 3x_2 + 5x_3 \leq 4000$ (availability of raw material A)

$4x_1 + 2x_2 + 7x_3 \leq 6000$ (availability of raw material B)

$x_1 \geq 200, x_2 \geq 200, x_3 \geq 150$, (Demand)

$2x_1 - 3x_2 = 0, 5x_2 - 2x_3 = 0$ (Market requirements)

$x_1 + \frac{x_2}{2} + \frac{x_3}{3} \leq 1500$ (Time constraint)

$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$.

Problem 10: *Show how the following objective function can be presented in equation form:*

$$\text{Minimize } Z = \max\{|x_1 - x_2 + 3x_3|, |-x_1 + 3x_2 - x_3|\}.$$

Solution: Let, $y = \max\{|x_1 - x_2 + 3x_3|, |-x_1 + 3x_2 - x_3|\}$.

Then, $|x_1 - x_2 + 3x_3| \leq y \Rightarrow x_1 - x_2 + 3x_3 \leq y, \quad -x_1 + x_2 - 3x_3 \leq y,$

And $|-x_1 + 3x_2 - x_3| \leq y \Rightarrow -x_1 + 3x_2 - x_3 \leq y, \quad x_1 - 3x_2 + x_3 \leq y.$

$$\text{Min } Z = y$$

Subject to:

$$x_1 - x_2 + 3x_3 \leq y$$

$$-x_1 + x_2 - 3x_3 \leq y$$

$$-x_1 + 3x_2 - x_3 \leq y$$

$$x_1 - 3x_2 + x_3 \leq y, \quad x_1, x_2, x_3 \geq 0.$$

Problem 11:

Two different products, $P1$ and $P2$, can be manufactured by one or both of two different machines, $M1$ and $M2$. The unit processing time of either product on either machine is the same. The daily capacity of machine $M1$ is 200 units (of either $P1$ or $P2$, or a mixture of both) and the daily capacity of machine $M2$ is 250 units. The shop supervisor wants to balance the production schedule of the two machines such that the total number of units produced on one machine is within 5 units of the number produced on the other. The profit per unit of $P1$ is \$10 and that of $P2$ is \$15. Set up the problem as an LP in equation form.

Solution: Let, x_{ij} be the number of units of product P_i from the machine M_j , $1 \leq i, j \leq 2$.

Total profit: $Z = 10(x_{11} + x_{12}) + 15(x_{21} + x_{22})$.

Therefore, total number of units produced by the machine M_1 : $x_{11} + x_{21}$.

Total number of units produced by the machine M_2 : $x_{12} + x_{22}$.

Objective function: $Max \quad Z = 10(x_{11} + x_{12}) + 15(x_{21} + x_{22}),$

Subject to:

$$x_{11} + x_{21} \leq 200$$

$$x_{12} + x_{22} \leq 250,$$

$$|(x_{11} + x_{21}) - (x_{12} + x_{22})| \leq 5$$

$$x_{11}, x_{12}, x_{21}, x_{22} \geq 0.$$

Problem 12:

A company produces two types of hats. Type 1 requires twice as much labour time as type 2 alone. If all labour time is dedicated to type 2 alone, the company can produce a total of 400 hats of type 2 per day. Respective market limits for two types are 150 and 200 per day. The profit is Rs. 8 per type 1 hat and Rs. 5 per type 2 hat. Find the optimum number of hats to be prepared so that it maximizes company's profits.

Solution: Let, x_1 number of Type 1 hat and x_2 number of Type 2 hats need to be prepared to maximize the profit.

Total profit

Subject to:

$$\text{Max: } Z = 8x_1 + 5x_2, \text{ (Objective function)}$$

$$x_1 \leq 150 \text{ (market limit)}$$

$$x_2 \leq 200 \text{ (market limit)}$$

$$\frac{x_1 t}{200} + \frac{x_2 t}{400} \leq t \Rightarrow 2x_1 + x_2 \leq 400, \text{ (time constraint)}$$

$$x_1, x_2 \geq 0.$$

References:

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- 2. H. S. Kasana and K.D. Kumar, Introductory Operations research, Springer publication, (2004)**
- 3. Chandra, S., Jayadeva, Mehra, A., Numerical Optimization and Applications, Narosa Publishing House, (2013).**

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